1 Cycle-slip detection and repair

The uncombined filter (the observation-equations section of the Foundations document) forms no linear combination in its measurement update. The per-signal $P^i_{r,s}$ and $\Phi^i_{r,s}$ enter the uncombined code and phase observation equations directly. Cycle slips are screened by a set of *diagnostic* combinations. In each one the geometric range ρ^i_r, the clocks $c(\mathrm{d}t_r - \mathrm{d}t^i)$ and the troposphere T^i_r cancel. What remains varies between epochs only by measurement noise and by any integer discontinuity of the carrier.

The status of the detectors is as follows.

- Melbourne–Wübbena (`src/core/slip/MwSlipDetector.cpp`, with its bodies split by concern over `MwSlipDetectorEpoch.cpp`, `MwSlipDetectorSample.cpp`, `MwSlipDetectorPairs.cpp`, `MwSlipDetectorStats.cpp`, `MwSlipDetectorQuery.cpp`, `MwSlipDetectorReset.cpp`, `MwSlipDetectorTrace.cpp` and `MwSlipDetectorZd.cpp`) is the primary detector and the only one armed by default.
- GRAPHIC (`src/core/slip/GraphicSlipDetector.cpp`) and the extra-wide-lane detector (`src/core/slip/EwlSlipDetector.cpp`), which needs a third frequency, are compiled and registered. Both are opt-in (Section 1.1), and the reference configurations (`config/golden/golden_field_refound.json` and `golden_sim_refound.json`) enable neither.
- The ionosphere-free phase detector (`src/core/slip/IfPhaseSlipDetector.cpp`) is *compiled and armed on the network path*. It is instantiated per station whenever `network.cycle_slip.if_phase_td` is set (Section 1.2.5), and both reference configurations set it. On the single-receiver user and kinematic path it stays staged (`src/core/slip/IfPhaseStagedUserPath.cpp`), because the position-uncertainty gate it needs is unsafe there.
- The geometry-free phase detector (`src/core/slip/GfSlipDetector.cpp`) is *staged*. It is in the tree but commented out of the explicit CMake source list, unlike the IF detector beside it. Its algebra is recorded here for completeness.

The user and kinematic set is composed at run time from the config block `estimation.cycle_slip`. The network composes its own per-station set from a parallel `network.cycle_slip` block (Section 1.1).

Throughout this section the receiver index r is suppressed (single-receiver PPP / PPP-RTK), and $f_1 > f_2$ denotes the two carriers of a band pair, sorted by frequency so that $f_1 - f_2 > 0$ and $\lambda_1 < \lambda_2$ (`src/core/slip/MwSlipDetectorPairs.cpp`, the band-frequency sort). Carrier phase is read from RINEX in cycles and scaled by $\lambda_s = c/f_s$ before combining, so Φ^i_s below is always in metres, consistent with the phase observation equation. The operator $\delta\,\cdot$ denotes a *temporal* integer slip step: for example δN_s is the change in the ambiguity on signal s across the slip epoch. It is distinct from the between-satellite single difference $\Delta\cdot$ of the differencing section of the Foundations document, which also uses Δ.

1.1 Detector registry and composition

The set of detectors is configurable. Each detector implements the interface `ISlipDetector` (`include/slip/ISlipDetector.hpp`), which has one entry point per epoch

$$\texttt{detect}(\text{SlipEpochContext}) \;\longrightarrow\; \{\,\text{SlipEvent}\,\},$$

plus lifecycle hooks (`resetSatellite`, `resetSatelliteSignal`, `enableTracing`, `setCorrections`). A `SlipEvent` is the union of the legacy per-detector result structs. Its `granularity` field distinguishes two kinds of flag. A *per-signal* flag (GRAPHIC, GF, IF) names one carrier. A *band-difference* flag (MW, EWL) reports a wide-lane integer $\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2$, which alone cannot assign the jump to one signal. Equation (21) resolves those two unknowns downstream. The inputs are bundled in `SlipEpochContext`, so every detector has the same signature. Only the IF detector uses the 3D position sigma and receiver ECEF it carries. On the network path, where that detector is armed, the position gate is always satisfied, because the estimator holds the station positions fixed.

`SlipDetectorRegistry::createEnabled` (`src/core/slip/SlipDetectorRegistry.cpp`) builds the set selected by `estimation.cycle_slip.{mw,graphic,ewl,gf,if}` in a fixed order. MW comes first. It is the primary detector and the only source of the wide-lane integers used by ambiguity resolution (the ambiguity-resolution document) and by the BSSD pivot logic (`src/core/filter/PivotManager.cpp`). The auxiliary detectors follow. MW defaults on and the auxiliaries default off, which keeps the historical MW-only behaviour.

A kind that is enabled in config but has no creator in the registry is skipped with a warning, so a config written for a later version still runs. Two kinds are skipped, for different reasons. The source file of `gf` is excluded from the build (Section 1.2.4). The source file of `if` *is* compiled, but the network path instantiates it directly (Section 1.2.5) and not through this registry, so `SlipDetectorRegistry` registers no `IonoFreePhase` creator.

`CycleSlipManager` drives the whole set through one loop. It reaches the MW detector's additional API (WL-lock queries, pivot-handover hints, per-satellite quality scoring) through a typed down-cast (`asMelbourneWubbena`), which keeps the base interface small.

The network does not use this registry for any detector. The `buildStation` lambda of `Application::networkSetup_` (`src/core/network/driver/ApplicationNetworkSetup.cpp`) constructs each station's set directly from a *parallel* config block, `network.cycle_slip.{mw_detect, turbo_edit, if_phase_td}`. These keys are distinct from their `estimation.cycle_slip` namesakes, and all three default off in the loader. Both reference configurations set `mw_detect` and `if_phase_td`; the loader also sets `mw_detect` when the phase-leveling carry is armed with no slip detector. `turbo_edit` stays off in the reference configurations (its setting is KEEP_OFF in `docs/CONFIG_promotion_ledger.json`). The two paths share the detector *classes*. They do not share the composition mechanism, the config block, or the response: the network refloats where the user path repairs (Section 1.3).

Composing by hand has one important consequence. With `network.cycle_slip.mw_detect` on and `network.cycle_slip.turbo_edit` off (and the phase-leveling carry armed, which the MW and TurboEdit detectors require), the station receives an `MwSlipDetector` and a null `TurboEditResolver`. The null pointer is what directs the detection branch away from integer repair and to the refloat path.

`turbo_edit` (Section 1.4) is *not* a detector kind. It is a *coordination mode*. When the MW detector is enabled, it pairs the MW wide-lane integer with the geometry-free phase jump to recover per-signal integers. It is documented separately from the detectors below.

1.2 Test-statistic combinations

1.2.1 Melbourne–Wübbena (wide-lane)

The Melbourne–Wübbena (MW) observable is the wide-lane carrier phase minus the narrow-lane code. Introducing the wide- and narrow-lane wavelengths

$$\lambda_{\mathrm{WL}} := \frac{c}{f_1 - f_2}, \qquad \lambda_{\mathrm{NL}} := \frac{c}{f_1 + f_2}, \tag{1}$$

the engine forms (`MwSlipDetector::computeMwMeters` in `src/core/slip/MwSlipDetectorPairs.cpp`)

$$\boxed{\mathrm{MW}^i \;=\; \underbrace{\frac{f_1\,\Phi_1^i - f_2\,\Phi_2^i}{f_1 - f_2}}_{\text{wide-lane phase}} \;-\; \underbrace{\frac{f_1\,P_1^i + f_2\,P_2^i}{f_1 + f_2}}_{\text{narrow-lane code}}.} \tag{2}$$

The range-additive terms common to the uncombined code and phase observation equations, namely geometry, clocks and troposphere, cancel between the two bracketed combinations. The slant ionosphere cancels too. The wide-lane phase has the ionospheric coefficient $-(f_1\mu_1 - f_2\mu_2)/(f_1 - f_2)$ and the narrow-lane code has $+(f_1\mu_1 + f_2\mu_2)/(f_1 + f_2)$. With $\mu_s = f_1^2/f_s^2$ (the ionosphere-scaling relation) the two are equal, so (2) removes the first-order ionosphere exactly. What remains is nearly constant, equal to the wide-lane ambiguity plus hardware bias,

$$\mathrm{MW}^i \;\approx\; \lambda_{\mathrm{WL}}\,N_{\mathrm{WL}}^i \;+\; b_{\mathrm{MW}}^i \;+\; \varepsilon_{\mathrm{MW}}, \qquad N_{\mathrm{WL}}^i = N_1^i - N_2^i, \tag{3}$$

so a wide-lane slip $\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2$ appears as a step of $\lambda_{\mathrm{WL}}\,\delta N_{\mathrm{WL}}$. For GPS L1/L2, $\lambda_{\mathrm{WL}} \approx 0.861\,\mathrm{m}$, and the dominant error is the narrow-lane *code* noise. With code noise $\sigma_P \sim 0.3\,\mathrm{m}$ the MW noise is $\sigma_{\mathrm{MW}} \approx \sigma_P/(\sqrt{2}\,\lambda_{\mathrm{WL}}) \sim 0.25$ wide-lane cycles, so the observable is limited by code noise. The integer-ready (BIA-corrected) variant subtracts the satellite MW bias b_{MW}^i, which has the same algebra as Equation (2) applied to the OSB phase and code biases of the uncombined code and phase observation equations.

The series $\mathrm{MW}^i(k)$ is tested against a robust adaptive gate. `estimation.cycle_slip.mw_detect_reference` selects the statistic.

The recommended choice is the first time difference $\mathrm{TD1}(k) = \mathrm{MW}^i(k) - \mathrm{MW}^i(k-1)$, gated against the median absolute deviation of its own history. The bundled real-time and replay configs set it explicitly. The second time difference $\mathrm{TD2}(k) = \mathrm{MW}^i(k) - 2\,\mathrm{MW}^i(k-1) + \mathrm{MW}^i(k-2)$ is available (`mw_td_order`). It is insensitive to curvature, at a noise penalty of $\sqrt{6}$ against $\sqrt{2}$ for TD1. `MwSlipDetector::activeStatScale` (`src/core/slip/MwSlipDetectorStats.cpp`) holds the white-noise scaling of each statistic.

TD1 is *not* the loader default. A config that leaves `estimation.cycle_slip.mw_detect_reference` unset uses the legacy Hatch mean, the deviation of the sample from the arc's running mean. It remains the default for backward compatibility and as a case for comparison, alongside the causal local median (`hatch_mean`, `local_median`, `time_difference`). In the Hatch and local-median modes the deviation is a level deviation. In the time-difference modes it is TD1 or TD2 itself, which is immune to a trend in the arc.

The threshold is derived from the data alone, and no σ from the stochastic model enters it. With $\hat{\sigma}$ the MAD-based spread of the active statistic over the arc,

$$\tau \;=\; k_{\mathrm{MW}}\,\max\!\big(\hat{\sigma},\; \sigma_{\min}\big), \tag{4}$$

where k_{MW} is `mw_sigma_multiplier` (default 8) and $\sigma_{\min}$ is the absolute floor `kMinSigmaMeters` ($0.01\,\mathrm{m}$). $\hat{\sigma}$ is used once the arc holds 15 samples (`kMinEpochsForAdaptive`). Before that, the floor is the larger of $\sigma_{\min}$, a young-arc MAD of the samples so far, and an EWMA-smoothed cross-satellite MAD of the zenith-normalised deviations of the other arcs, re-expanded by $1/\sin\theta$ (the elevation clamped at 10° from below). The cross-satellite pool widens in steps when a group is sparse: constellation and band pair, then the constellation, then all satellites except GLONASS. With no pool at any tier, the gate is a gross-slip bound of $\min(\lambda_{\mathrm{WL}}, 1\,\mathrm{m})$. The construction is in `MwSlipDetector::computeAdaptiveThreshold` (`src/core/slip/MwSlipDetectorStats.cpp`) and its empirical constants are not reproduced here.

Wide-lane lock. The detector also supplies the wide-lane integers that ambiguity resolution consumes. An arc keeps the running mean $\bar{m}$ of MW^i and its standard error $\sigma_{\bar{m}}$. With $\bar{n} = \bar{m}/\lambda_{\mathrm{WL}}$ and $f = \bar{n} - \mathrm{round}(\bar{n})$, the arc locks to the integer $\mathrm{round}(\bar{n})$ when it holds at least 10 samples and

$$|f| + 3\,\sigma_{\bar{m}}/\lambda_{\mathrm{WL}} \;<\; \tfrac{1}{2}, \qquad |f| < 0.30, \tag{5}$$

and it is demoted when the running mean drifts more than 0.4 cycle from the locked integer (`kWlLockDriftMaxCyc`). Where a satellite bias product (BIA) is applied, $\bar{m}$ is the bias-corrected statistic. Otherwise, an arc with at least 20 samples, $|f| \le 0.20$ and $\sigma_{\bar{m}}/\lambda_{\mathrm{WL}} \le 0.05$ may commit its residual $f\,\lambda_{\mathrm{WL}}$ as an in-house satellite bias, blended across arcs by an EWMA of weight 0.2 and refused when it disagrees with the stored value by more than 5σ (`MwSlipDetector::tryCommitOrUpdateBias`).

1.2.2 Extra-wide-lane (EWL)

For satellites with three or more frequencies the MW algebra (Equation (2)) is applied to a pair of *non-L1* carriers (`EwlSlipDetector::computeEwlMwMeters`; band digit ≥ 2). The two carriers are close in frequency, so $f_1 - f_2$ is small and the extra-wide-lane wavelength

$$\lambda_{\mathrm{EWL}} \;=\; \frac{c}{f_1 - f_2} \tag{6}$$

is very long: $\lambda_{\mathrm{EWL}} \approx 5.86\,\mathrm{m}$ for GPS L2/L5 and $\approx 9.77\,\mathrm{m}$ for Galileo E5a/E5b. Metre-level code noise is then $\lesssim 0.05$ EWL cycles. The EWL integer $N_{\mathrm{EWL}} = N_2 - N_5$ resolves in one epoch, and the size of a slip δN_{EWL} is determined almost without error. With the L1/L2 wide-lane (Equation (3)) this gives two independent band-difference constraints,

$$\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2, \qquad \delta N_{\mathrm{EWL}} = \delta N_2 - \delta N_5, \tag{7}$$

which leave the per-signal triple $(\delta N_1, \delta N_2, \delta N_5)$ with one degree of freedom [2]. The detector tests the deviation of the EWL statistic from the arc mean against $\max(k_{\mathrm{EWL}}\hat{\sigma},\ \tfrac{1}{2}\lambda_{\mathrm{EWL}})$, with k_{EWL} the `ewl_sigma_multiplier` (default 5) and $\hat{\sigma}$ the MAD-based spread of the between-epoch differences over a 30-epoch window; the floor is half an extra-wide-lane cycle.

1.2.3 GRAPHIC (single-frequency half-sum)

The GRAPHIC combination averages the code and phase on a *single* band (`GraphicSlipDetector::computeGraphic`):

$$\mathrm{GR}_s^i \;=\; \tfrac{1}{2}\big(P_s^i + \Phi_s^i\big), \tag{8}$$

with Φ_s^i in metres. The slant ionosphere enters as $+\mu_s I_{r,1}^i$ on the code and $-\mu_s I_{r,1}^i$ on the phase of the *same* signal (the uncombined code and phase observation equations), so the half-sum cancels it to first order. GRAPHIC is therefore ionosphere-free with one frequency, but it keeps half the code noise. The combination still contains the geometric range, the clocks, the troposphere, and the constant half-ambiguity $\tfrac{1}{2}\lambda_s N_s^i$. The between-epoch first difference

$$\mathrm{dGR}_s^i(k) \;=\; \mathrm{GR}_s^i(k) - \mathrm{GR}_s^i(k-1) \tag{9}$$

removes the constant ambiguity but not the geometry: dGR still holds the range rate times the epoch spacing, hundreds of metres for a moving satellite. Two tests are implemented, selected by `estimation.cycle_slip.graphic_trend_centred` (default off, setting status UNVALIDATED).

The default compares |dGR| with the gate. The gate is a multiple of the MAD sigma of the recent differences, a spread about their median, while |dGR| is measured from zero. For a moving satellite the test therefore fails at the first epoch the window gives a gate (the sixth epoch of an arc) and resets the arc there, again and again. It is kept as the default so that a run that selects the detector keeps its output. The integer estimate is $\delta N = \mathrm{round}\big(2\,\mathrm{dGR}/\lambda_s\big)$.

The trend-centred test uses the change of the difference from one epoch to the next, measured from its median,

$$r_s^i(k) \;=\; \mathrm{dGR}_s^i(k) - \mathrm{dGR}_s^i(k-1) \;-\; m_s^i(k), \tag{10}$$

where $m_s^i(k)$ is the median of that change over the last 30 epochs of the arc. A constant range rate cancels in the change, and the slowly varying range acceleration is the median, so a moving satellite sits at zero and only a jump or noise remains. The gate is a multiple of the MAD sigma of the same window, so the threshold and the tested value share one centre. A slip of δN cycles on signal s produces a jump of $\frac{1}{2}\lambda_s\,\delta N$ in r, and the integer estimate is $\delta N = \mathrm{round}\big(2\,r/\lambda_s\big)$. The first possible detection is one epoch later than in the default, since the second difference exists from the third epoch of an arc. With $\sigma_P \sim 0.3\,\mathrm{m}$ the noise of r is $\sim 0.4\,\mathrm{m}$, so the 6σ gate sits near 2 m, about 20 L1 cycles. The hard floor `kMinThresholdM` $= 0.5\,\mathrm{m}$ (≈ 5 L1 cycles) applies only on very quiet arcs. GRAPHIC is therefore a coarse backstop and not a sub-cycle detector [1, 3].

1.2.4 Geometry-free phase (staged)

The geometry-free (GF) observable is the between-band phase difference

$$\mathrm{GF}^i \;=\; \Phi_1^i - \Phi_2^i \;\approx\; (\mu_2 - \mu_1)\, I_{r,1}^i \;+\; (\lambda_1 N_1^i - \lambda_2 N_2^i), \tag{11}$$

which removes geometry, clocks and troposphere exactly but keeps the *difference* of the two ionospheric delays. $I_{r,1}^i$ varies smoothly, so a second-order time difference

$$\mathrm{TD2}_{\mathrm{GF}}(k) \;=\; \mathrm{GF}^i(k) - 2\,\mathrm{GF}^i(k-1) + \mathrm{GF}^i(k-2) \tag{12}$$

cancels the ionospheric trend up to quadratic order over the sampling interval and leaves noise plus any slip (`GfSlipDetector::computeGfMeters` for Equation (11); the TD2 update is the test statistic for a level jump). The detector needs a high data rate ($\geq 1\,\mathrm{Hz}$). At a coarse rate the remaining ionospheric curvature is too large for (12). The detector is in the tree but excluded from the build.

The TD2 statistic (12) is still used elsewhere. `slip/IonoVariabilityMonitor` forms the same second difference per satellite on the two lowest RINEX bands (`selectPrimaryPair`, `computeGfMeters`), and it *is* compiled. It is a process-noise *monitor*, not a detector. Its per-satellite output is the clamped squared ratio of two MAD-derived values of σ for $\mathrm{TD2}_{\mathrm{GF}}$, one over a recent window and one over a baseline window,

$$m^i \;=\; \mathrm{clamp}\!\left(\left(\hat{\sigma}_{\mathrm{recent}}/\hat{\sigma}_{\mathrm{baseline}}\right)^2,\;\; 1,\;\; m_{\max}\right), \tag{13}$$

with a small floor on $\hat{\sigma}_{\mathrm{baseline}}$. The ratio normalises itself. The slowly varying obliquity and curvature content of Equation (11) is present in *both* windows and cancels. A quiet sky gives $m^i \approx 1$ at any elevation, and the $1/\sin^2\theta$ obliquity scaling already in the ionospheric process noise is not counted twice. The monitor treats a cycle slip as an isolated single-epoch outlier, which the MAD is designed to *reject*. That is the opposite of a detection statistic, so the monitor is not a substitute for the staged detector.

The monitor serves the user path only, and `estimation.adaptive_iono_process_noise.enable` (default off) arms it: m^i scales the static ionospheric spectral density into a per-satellite zenith override, which `StateManager::setIonoProcessNoiseOverrides` consumes. The real-time engine does not use it, and the network reads no monitor output (the network setting that once did was deleted). The window lengths and the clamp $m_{\max}$ are empirical and stay in config.

1.2.5 Ionosphere-free phase (staged on the user path; armed on the network path)

The dual-frequency ionosphere-free (IF) phase combination uses the coefficients

$$c_1 = \frac{f_1^2}{f_1^2 - f_2^2}, \qquad c_2 = \frac{f_2^2}{f_1^2 - f_2^2}, \qquad c_1 - c_2 = 1, \tag{14}$$

giving $\Phi_{\mathrm{IF}}^i = c_1 \Phi_1^i - c_2 \Phi_2^i$ and the effective wavelength $\lambda_{\mathrm{IF}} = \left| c_1 \lambda_1 - c_2 \lambda_2 \right|$ (`IfPhaseSlipDetector::computeIfWavelengthMeters`); for GPS L1/L2 $\lambda_{\mathrm{IF}} \approx 0.107\,\mathrm{m}$. The detector does not test the raw combination. It forms the IF observed-minus-computed (OMC) value and differences it twice. The first difference is between satellites, against a high-elevation pivot p_g. It removes the receiver clock, the inter-system offset $c\,\delta t_r^{(g)}$ and the satellite phase clock. The second is between epochs. It removes the slowly varying wet troposphere and the constant ambiguity:

$$\nabla\Delta\Phi_{\mathrm{IF}}^{i,p_g}(k) = \left[\Phi_{\mathrm{IF}}^i(k) - \Phi_{\mathrm{IF}}^{p_g}(k)\right] - \left[\Phi_{\mathrm{IF}}^i(k-1) - \Phi_{\mathrm{IF}}^{p_g}(k-1)\right], \tag{15}$$

which is near zero for a clean arc and jumps by $\sim \delta N\, \lambda_{\mathrm{IF}}$ on a slip. In (15) the symbol Φ_{IF} stands for the observed-minus-computed value, since the model subtraction is what removes the satellite range and clock. The between-satellite step does not cancel the differential geometric error $(\boldsymbol{e}^i - \boldsymbol{e}^{p_g})^{\mathrm{T}} \delta\boldsymbol{r}$, so flagging must be gated by the filter's 3D position sigma. The gate requires a static site with a precise a priori position. The single-receiver kinematic path cannot guarantee that. There the detector is never instantiated, and the user toggle `estimation.enable_if_slip_detector` has no effect. The network estimator holds every station position *fixed*, so the gate is always satisfied, and the detector is armed there.

On the network path (the `buildStation` lambda of `Application::networkSetup_`, gated on `network.cycle_slip.if_phase_td`) an `IfPhaseSlipDetector` is instantiated per *non-pivot* station. The pivot's ambiguities are fixed by the gauge, so on the legacy datum path the pivot has nothing to refloat. Under the prioritised phase datum `network.phase_datum_prioritized`, which the reference configurations use, the detector is instantiated for every station, because the former pivot then has arcs that are not anchored and still need detection. By default the detector forms one signal pair per satellite (the first two distinct phase bands in key order). With `network.cycle_slip.if_phase_td_pairs` set to `all`, which the reference configurations do, it forms every pair of distinct frequencies, each on its own between-epoch gradient chain, pool and reset key, so a slip that leaves the first pair silent is still tested on the others. The detector is built with `setDeferFlagging(true)`. It caches each per-epoch $\nabla\Delta\Phi_{\mathrm{IF}}^{i,p_g}$ of Equation (15) and does *not* apply a threshold locally. The driver (`networkSlipIfTdDriver` in `src/core/network/slip/SlipQc.cpp`) *pools* the deferred jumps across stations by signal pair and removes their median med. The median is the network common mode, a wobble of the frame or datum. It does not depend on elevation, so it is removed in raw metres. Each residual is then normalised to a zenith equivalent,

$$z \;=\; \left(\nabla\Delta\Phi_{\mathrm{IF}}^{i,p_g} - \mathrm{med}\right) \sin\theta, \tag{16}$$

and the pooled robust scale $\hat{s} = 1.4826\,\mathrm{median}\,|z|$ is formed (`ifTdPoolStats`). The reference satellite of each constellation is left out of the scale, because its double difference is zero by construction

and would deflate the median. At each satellite's threshold the scale is multiplied by $1/\sin\theta$ again, so *one* scale serves the whole sky. The map is the free function `ifElevMap` ($1/\sin\theta$, clamped at 5° so that it stays bounded near the horizon, which is already elevation-masked). The effective scale $\hat{s}_{\text{eff}}$ is an empirical-Bayes blend of the pooled prior with the station pair's own EWMA scale (`ifTdShrunkScale`). The EWMA carries the weight $w = n/(n+\kappa)$, with n the number of clean samples of the station pair. A quiet reference station and a noisy station with modernised signals therefore each get their own envelope, and the majority does not set one threshold for all. A satellite at elevation θ is flagged when

$$\left|\nabla\Delta\Phi_{\text{IF}}^{i,p_g} - \text{med}\right| \;>\; \max\!\left(\frac{k\,\hat{s}_{\text{eff}}}{\sin\theta},\;\; c_{\text{floor}}\,\lambda_{\text{IF}}\right), \tag{17}$$

where k is a loose MAD multiplier. The floor is proportional to the $(1,1)$ IF jump $\lambda_{\text{IF}} = c/(f_1+f_2) \approx 10.7\,\text{cm}$ and is kept in raw metres, so a simultaneous one-cycle slip is caught at any elevation. Only the clean survivors are folded into the station-pair EWMA, so the scale does not absorb the event it has to catch. The settings k, c_{floor}, κ and the EWMA weight are `network.cycle_slip.if_phase_td_k_mad` (default 8), `if_phase_td_floor_frac` (1.0), `if_phase_td_shrink_kappa` (20) and `if_phase_td_ewma_alpha` (0.15). A fifth setting, `if_phase_td_min_pool`, only counts pools below a size for a log line and enters no decision. They are read once per call inside the driver.

A flag never repairs and never resets filter state in place. It restarts the detector's own arc (arc hygiene). Its two estimator-side consequences are gated separately inside `refloat_pair`. The flagged pair enters the slip funnel of Section 1.3.1 when the double-difference invariant carry is armed. The band ambiguities are marked pending-reseed (the refloat itself) only when the phase-leveling carry is warm. On a cold or absent carry the flag is emitted as diagnostic telemetry and changes nothing, so a clean run log is no evidence that the response acted; the run-end counters are. The loose gate is intentional. The driver is a coarse first pass, and the sharp integer decision is left to a post-estimation phase validation that is not yet built; it is not the discriminating solver of Section 1.4. What the network does with a flag (which bands it frees, how the level is re-established) is the subject of the event-response document.

A restart leaves the detector with no double difference for that satellite at the next epoch, so a slip repeated one epoch later passes unseen. With `network.cycle_slip.if_phase_td_reseed_on_flag` (default off, setting status UNVALIDATED) the chain instead keeps the flag epoch's between-satellite gradient as the origin of the next double difference (`IfPhaseSlipDetector::reseedSatellite`). The next double difference then measures only what happened between the two epochs: about zero on an arc that stays clean, and the size of a second step if the slip repeats. A one-epoch spike reads with the opposite sign at the next epoch, so under the re-seed it flags twice where a restart flags once.

The reference satellite. The pool cannot see a slip on the between-satellite reference itself. Such a jump is common to every satellite of that station's chain and is removed with med, and the reference's own double difference is zero. The detector therefore tests the reference separately. A slip on the reference appears as the same step on all n non-reference double differences d_j of the chain, so their median $\text{med}_j\, d_j$ estimates it and their spread $s_d = 1.4826\ \text{median}_j|d_j - \text{med}_j\, d_j|$ measures the noise. The legacy gate fires on $|\text{med}_j\, d_j| > \frac{1}{2}\lambda_{\text{IF}}$ with at least two non-reference members. The empirical gate (`network.cycle_slip.if_phase_td_ref_gate`, default off, set in both reference configurations) keeps that condition as the candidate filter and decides each candidate by three further conditions.

1. Scale: $|\text{med}_j\, d_j| > \max(\phi\,\lambda_{\text{IF}},\; k_r\,\hat{s}_r)$, with $\hat{s}_r$ an empirical-Bayes blend of the chain's EWMA of s_d with the cold prior $\phi\,\lambda_{\text{IF}}/k_r$, $k_r = 6$ and $\phi = 0.5$ (`if_phase_td_ref_k`, `if_phase_td_ref_`

floor_frac). A chain with fewer than 4 non-reference members widens the physical floor by a factor of 3.

2. Coherence: a genuine reference slip is a common step, so $s_d \leq 0.5\,|\mathrm{med}_j\, d_j|$. When the family is incoherent and at least $\max(2, \lceil n/2 \rceil)$ members individually exceed λ_{IF}, the event is a multi-satellite station event and the reference is reset without the median rebase. Any other incoherent family is a drift and does not fire.
3. Persistence: after more than one consecutive firing on the same reference (`if_phase_td_ref_max_consecutive`, default 1) the gate stops firing and forces a re-election that excludes that satellite, which then enters the pool test.

The reset keys are parsed by `parseIfPairResetKey` and the flagged pair is refloated through the same `refloat_pair`. Both bands of the pair are refloated, because the IF combination cannot localise which one slipped. The exception is the common-band scope of Section 1.3, which, when armed, keeps the bands the evidence exonerates. With `if_phase_td_ref_arm_all` the reference test also runs on the additional chains of the `all` pair mode. Two further refinements, an abstention while the empirical scale has fewer than κ samples and an elevation-normalised bar, are UNVALIDATED and off. The gate is pushed only on the network path; the staged user and kinematic path keeps the legacy fixed gate.

1.3 Network slip QC: raw evidence and refloat over repair

The IF-phase driver of Section 1.2.5 is one component of the network's slip layer. The components are free functions in `network/slip/SlipQc` (`src/core/network/slip/SlipQc.cpp`), called in a fixed serial, station-ordered sequence from the per-epoch PASS A slip scan (`Application::networkPassASlipScan_`). The scan is an ordered list of named stages, declared in `network/slip/PassASlipScan` (the detection half) and `network/slip/PassASlipDrains` (the response half), and each stage reads the epoch through one view of references (`PassAView`).

1. The parallel station scan (`passAScanStations`) models each station and runs its MW and IF-phase detectors. It writes only index-addressed per-station slots, so it is safe under OpenMP. Under `mw_detect` it also collects the admitted refloat candidates (`passAMwAdmit`) and the evidence for the common-band scope (`passAMwEvidence`).
2. The serial folds: station presence, drop reasons, the second code and second phase channel bookkeeping.
3. The shared raw observables (`passARawObservables`), below.
4. The MW admission counts, which are formed whether or not any setting acts on them.
5. The common-band verdict (`passACommonBandVerdict`), formed once before any drain.
6. The drains, in order: the integer-repair drain (`networkSlipDrainTORepairs`, applying the integer repairs of the network TurboEdit resolver to the phase-leveling carry), the MW and LLI drain (`networkSlipDrainMwRefloat`), the IF-phase driver, the arc-scoped geometry-free test (Section 1.3.2), the wind-up restart drain, the two gap re-entry drains (per satellite and per signal) and the evaluation-only injected-free drain, each behind its own setting (off in the loader).

The shared evidence is a separate component, `network/slip/RawObservables`.

Shared evidence. Once per epoch per (station, satellite), `networkComputeRawObservables` forms the MW combination of Equation (2) and the GF combination of Equation (11) on the two lowest distinct bands whose phase *and* matching code are both present. It also records the first band's phase-minus-code level and the per-band RINEX LLI byte. These are absolute levels, not differences. `networkQualifyRawObservables` then adjudicates each observable's step against the immediately adjacent processed epoch, that is $\mathrm{MW}^i(k) - \mathrm{MW}^i(k-1)$ and $\mathrm{GF}^i(k) - \mathrm{GF}^i(k-1)$. The threshold has the shape of Section 1.2.5: a multiple of an empirical scale, floored from below,

$$g_{\mathrm{MW}} = \max\bigl(k_{\mathrm{mw}}\,\hat{s}_{\mathrm{MW}},\ 0.5\,\lambda_{\mathrm{WL}}\bigr), \qquad g_{\mathrm{GF}} = \max\bigl(k_{\mathrm{gf}}\,\hat{s}_{\mathrm{GF}},\ 0.04\,\mathrm{m}\bigr), \tag{18}$$

with $k_{\mathrm{mw}} = k_{\mathrm{gf}} = 5$. The empirical term $\hat{s}$ is a per-station robust EWMA (weight 0.15) of that station's own clean steps, built from the same $1.4826\,\mathrm{median}\,|\cdot|$ estimator (exposed for test as `networkRobustStepScale`); a station with no folded history takes the scale of the epoch's own steps as its prior. The floor is a physical integer bound kept in raw metres: a fraction of a wide-lane cycle λ_{WL} for MW (default 0.5 cycle) and a few centimetres for GF (default 4 cm). The gate therefore falls back to the physical bound when the empirical term is cold or degenerate, never to a stochastic-model σ. The MW adaptive threshold of Section 1.2.1 follows the same rule. The ordering also matches the IF-TD pool arm: each satellite is thresholded with the pre-update scale, and only the clean survivors are folded into the EWMA, so the scale does not absorb the event it has to catch. A step is measurable only against an adjacent predecessor. Across a gap it cannot be measured, and the code treats that absence of evidence as neither evidence of a slip nor evidence of continuity. The multipliers and floor values live in `SlipQcParams`.

MW admission. A flag of the detect-only MW path becomes a refloat candidate when it passes either of two rules (`networkMwAdmitCandidate`, a pure function). The *integer rule* requires a known non-zero wide-lane integer whose rounding residual is within `network.cycle_slip.turbo_edit_frac_max` (default 0.15 cycle). This rule grades how cleanly the step rounds and says nothing of how far the step stands above the noise, although the detect-only path never applies the integer. The *evidence rule* (`network.cycle_slip.mw_evidence_refloat`, off in the loader, set in both reference configurations) tests the second property directly: the lane has no satellite bias correction, the arc holds at least `mw_evidence_min_arc_epochs` (10) samples, and |deviation|/threshold $\geq$ `mw_evidence_ratio_min` (3.0). Bias-corrected lanes are excluded, because there a re-reference of the detector's own pivot appears as a synchronous step on several satellites. Candidates that fail both rules are counted by cause (loose residual, unknown integer), with the same counts formed when the evidence rule is off, so the effect of arming it is sized beforehand.

Common-band scope. On a triple-frequency arc, a slip on one band moves every MW pair that contains the band, and each firing pair would free both of its bands, so the union is all three. The scope (`network.cycle_slip.common_band_scope`, set in both reference configurations) decides per (station, satellite, epoch) which bands the evidence implicates, by set algebra. An MW pair is *silent* when the detector processed a sample for it, its arc held at least `common_band_min_arc_epochs` (10) samples and it did not flag. A silent pair exonerates its two bands, unless its IF-phase pair flagged this epoch (the equal-count slip leaves MW silent while the IF combination moves λ_{IF} per cycle), unless the IF reference test cannot test it (`common_band_ref_arc`: on the reference satellite the double difference is zero by construction), and unless one of its bands carried a loss-of-lock bit at the previous epoch. A band with a loss-of-lock bit is implicated regardless. With B the tracked bands,

$$\text{implicated} \;=\; B \setminus \bigl\{b:\ \text{no LLI on } b \ \wedge\ \exists \text{ silent, un-vetoed pair } p \ni b\bigr\}. \tag{19}$$

Narrowing needs positive exoneration: with no MW flag, nothing exonerated, or an empty implicated set, the response stays all-band. The verdict is formed once per epoch before the drains and governs each of them, so the per-epoch union of freed bands is exactly the implicated set. The scope only narrows a response; the responses themselves are described in the event-response document.

Code-bias jump test. The emitted code biases carry a per-signal discontinuity counter that the user receiver reads. The structural triggers (a clock-pivot change, a re-reference of a signal group, a change of a satellite's band set) detect gauge re-references but not an estimate that wanders under weak separation. With `network.code_bias_disc_jump_test` (off in the loader, set in both reference configurations) the counter also steps when the emitted code bias of a satellite and signal differs from its previous emission by more than the phase-bias threshold of 0.05 m, provided the baseline is fresh. It is a magnitude test on the emitted product and reads no observation; the product and its discontinuity declarations belong to the emitted-product document.

Response doctrine. On the network path a detection refloats: it marks the ambiguity pending-reseed so that PASS B re-estimates it from that epoch's data, and it never applies an integer. This differs from the user-path TurboEdit repair of Section 1.4. The reason is the network's missing satellite phase biases. Without them the network wide-lane integer is not trustworthy, and a wrong integer written into a carried value would bias the gauge, whereas a refloat costs only re-convergence. The one exception is `networkSlipDrainT0Repairs`, active only when the network TurboEdit key of Section 1.4 is armed. It shifts the phase-leveling carry by $\delta N_s\,\lambda_s$ continuously, with the variance unchanged and the covariance preserved. It does so only for an integer that is both confident and of sane magnitude. Both bounds are network settings, separate from the resolver's own constants: a rounding residual (Equation (26)) of at most 0.15 cycle and a magnitude of at most 50 cycles by default. A low-confidence detection is left untouched: it is neither repaired, refloated nor dropped. The float wide-lane's low-confidence repairs are measured to be mostly false positives, and acting on them discards a good carry. The resolver's reset set is not acted on either; without satellite phase biases it flags tens of satellites per epoch. The refinements of the response (the MW re-flag skip, which frees no band twice for one step, the co-resumed share of a gap re-entry, and the re-levelling of freed arcs) are described in the event-response document.

Status. Every component is off in the loader, and the reference configurations enable the following. `network.cycle_slip.mw_detect` enables the MW/LLI drain, `network.cycle_slip.mw_lli_corroborate` the LLI qualification and `network.cycle_slip.gap_instrument` the arc-resumption instrument. The raw observables are computed only when a reader asks for them (`SlipQcParams::anyConsumer`). The adjudication above runs only when a decision setting is armed; a run with the raw trace alone computes the levels and adjudicates nothing. Every use of the verdicts in the drains is therefore guarded on whether the evidence exists this epoch. Without it the LLI path stays decisive, because a gate that fired on missing data would do more harm than the chatter it is meant to suppress. The qualification never excludes an LLI assertion. An assertion acts when a measured MW or GF step corroborates it, or when the station's own LLI behaviour is still judged credible by a two-state hysteresis on the EWMA of its LLI-set fraction (`networkLliCredibilityStep`). Only an uncorroborated assertion at a station whose LLI is judged not credible is downgraded. It is then emitted as suppressed and counted, not dropped.

The raw step evidence has a second reader outside this chapter. With `network.carry_invariant_innov_corroborate` armed (off in the loader and in the reference configurations, setting status KEEP_OFF), the innovation gate on the carried double differences does not re-tap a chord

whose arc the raw observables measured as continuous. The gate is fail-safe in one direction only: a measured flat arc suppresses the re-tap, and absence of evidence still re-taps. The slip response is therefore narrowed and never removed. The gate is part of the estimator and is described in the between-epoch carry document.

1.3.1 The slip funnel

The epoch's *slip funnel* (`NetworkSlipFunnel` in `network/slip/SlipFunnel`) is the set of (station, ambiguity) arcs that the epoch treats as slipped. It is the one list the carried quantities' slip responses read, so a detector adds an arc to it once and every reader sees the same set. It has two members: `carry_slipped_ambs`, the funnel proper, and `carry_slipped_raw`, the subset a raw detector saw step. It is empty at the start of an epoch, filled before the phase datum is elected, read from the datum on, and gone at the epoch's end. A flag that must survive to the next epoch is run state, never a funnel entry.

The producers, in epoch order, each take the funnel as an output.

1. `networkSlipFunnelOpen`: the previous epoch's flags from the innovation gate, consumed one-shot.
2. The PASS A drains of the preceding subsection (MW and LLI, IF-phase, geometry-free arc test, wind-up restart, gap re-entry), into the raw subset.
3. `networkSlipFunnelUnionRaw`: the raw subset joins the funnel.
4. The signal-integrity and measurement-admission refloats, into both sets.

The readers are the phase-anchor election (anchor hygiene, the chain datum, the anchor-slip handover), the gap-resumption count (`networkGapResumeCensus`; a marked arc is never counted as bridged), the erase policy of PASS B, the double-difference carry (a funnel arc leaves the carried set and is re-tapped fresh) and the emission. The funnel serves two design rules. A stiff quantity carried over time must have a slip response, and the funnel is the feed of that response. A mechanism that changes which information reaches which states is an event at both edges, with counters and identities; each funnel entry carries its identity and is the decoupling edge of that arc for the epoch, while the re-coupling edge is the gap-resumption count and the arc ledger.

1.3.2 Arc-scoped geometry-free jump test

Three slip classes pass the MW, IF time-difference and loss-of-lock detectors: an equal count on both bands (the IF step sits on the IF floor and MW does not move), a family of one or two wide-lane cycles whose geometry-free step the free slant ionosphere absorbs, and a slip repeated at the epoch after an IF flag. The raw geometry-free phase step is the one pre-fit witness of all three. `network.cycle_slip.gf_arc_jump` (off in the loader, setting status UNVALIDATED) tests it per (station, satellite) dual-frequency arc (`network/slip/GfArcJump`). With the geometry-free level $\mathrm{GF}(t)$ of (11) on the primary pair, its one-epoch step $d(t) = \mathrm{GF}(t) - \mathrm{GF}(t-1)$ and t_c the arc's last *clean* step,

$$r(t) = d(t) - d(t_c), \qquad \text{flag when} \quad |r(t)| > \max\big(\gamma,\ k_{\mathrm{jump}}\, 1.4826\, \mathrm{MAD}(\text{recent clean } r)\big). \tag{20}$$

The floor γ is `gf_arc_jump_m` (0.04 m, below the 0.0539 m geometry-free step of an equal one-cycle slip on GPS L1/L2), and the scale term is off at the default `gf_arc_jump_k` of 0, the multiplier k_{jump}; it applies once an arc holds `gf_arc_jump_min_count` (10) clean values of the last `gf_arc_jump_window` (20). The reference is the last clean step, not the previous one, so a flagged step never

becomes the reference and a slip repeated at the next epoch still differs from it, while an isolated slip flags once. An arc is consecutive processed epochs on one pair with one step span; a gap, a pair change or a span change restarts it, and testing begins at its third sample. After two consecutive flags the reference is re-seeded from the next step, which bounds the flags that a persistent change of the ionospheric rate can cause. A flag frees the pair's two bands as an IF-phase flag does, subject to the common-band verdict.

1.4 TurboEdit dual-frequency resolution

The TurboEdit resolver is a pure MW+GF joint integer solver (`include/slip/TurboEditResolver.hpp`). Its bodies are split by concern: the per-epoch driver, construction and statistics in `src/core/slip/TurboEditResolver.cpp`, the first pass over the observations (the geometry-free chains and the cross-satellite pools) in `TurboEditChain.cpp`, the adjudication of each row and the reconciliation of the epoch in `TurboEditAdjudication.cpp`, and the gap bridge in `TurboEditBridge.cpp`. It owns the per-arc geometry-free jump history and the integer arithmetic. It returns per-signal integer repairs and a reset-fallback set, and it does not modify filter state. It is wired into the post-processing `CycleSlipManager` and the real-time engine, and is enabled by `estimation.cycle_slip.turbo_edit`.

That key is default off. The bundled real-time and SBF-replay configs turn it on; the internal PPP-RTK AR config leaves it off. The network has a separate key, `network.cycle_slip.turbo_edit`, which builds one resolver per station and feeds the repair drain of Section 1.3. On the network the integer repair is licensed only at about 1 s sampling, where the geometry-free drift between epochs is small; the loader steers 30 s runs to `mw_detect`. With that key off and `network.cycle_slip.mw_detect` on, the same MW detector runs as a pure detector, and the network refloats instead of repairing (Section 1.3). Of the sub-toggles below, the soft-LLI bridge (`turbo_edit_bridge_lli`) is default off and the MW-only code-domain veto (`turbo_edit_mw_only_veto`) is default on.

1.4.1 The 2×2 system

For a dual-frequency phase pair (s_1, s_2) on satellite i, ordered higher frequency first ($f_{s_1} > f_{s_2}$, hence $\lambda_{s_1} < \lambda_{s_2}$), the resolver pairs two scalar slip observables. The first is the MW wide-lane integer of Equation (3). It is taken as input from the MW detector and never recomputed, so the empirically tuned MW path of Section 1.2.1 is the single source of δN_{WL}:

$$\delta N_{\mathrm{WL}} \;=\; \delta N_{s_1} - \delta N_{s_2} \qquad \text{(one equation, two unknowns).} \tag{21}$$

The second is the GF phase jump in metres. Writing the GF level as $\mathrm{GF}(t) = \lambda_{s_1} L_{s_1}(t) - \lambda_{s_2} L_{s_2}(t)$ (carrier in cycles), a simultaneous slip steps its ambiguity content by

$$\delta\mathrm{GF} \;=\; \lambda_{s_1}\,\delta N_{s_1} - \lambda_{s_2}\,\delta N_{s_2}, \tag{22}$$

estimated within the arc by the second-order time difference $\mathrm{TD2}(t) = \mathrm{GF}(t) - 2\,\mathrm{GF}(t-1) + \mathrm{GF}(t-2)$ of Equation (12). TD2 cancels the ionospheric delay up to quadratic order, so at the slip epoch it equals the integer jump plus phase noise. The TD2 chain requires three consecutive epochs; a non-consecutive history re-seeds the chain (Section 1.4.4).

Stacking Equations (21) and (22) gives the linear map from the integer slip pair to the two observed jumps:

$$\boxed{\begin{pmatrix} \delta N_{\mathrm{WL}} \\ \delta\mathrm{GF} \end{pmatrix} \;=\; \underbrace{\begin{pmatrix} 1 & -1 \\ \lambda_{s_1} & -\lambda_{s_2} \end{pmatrix}}_{\boldsymbol{A}} \begin{pmatrix} \delta N_{s_1} \\ \delta N_{s_2} \end{pmatrix}.} \tag{23}$$

The determinant $\det \boldsymbol{A} = \lambda_{s_1} - \lambda_{s_2}$ is non-zero for any cross-band pair. It is the centimetre-scale quantity `denom` of Equation (25), $\approx -0.0539\,\mathrm{m}$ for GPS L1/L2, and is distinct from the $\sim 0.86\,\mathrm{m}$ wide-lane wavelength. The system is therefore always invertible:

$$\boldsymbol{A}^{-1} = \frac{1}{\lambda_{s_1} - \lambda_{s_2}} \begin{pmatrix} -\lambda_{s_2} & 1 \\ -\lambda_{s_1} & 1 \end{pmatrix}. \tag{24}$$

The implementation does not invert the full matrix. It solves the second row for δN_{s_2} and back-substitutes into the first (`TurboEditAdjudication.cpp`, and `TurboEditBridge.cpp` for the bridged jump):

$$\delta N_{s_2} \;=\; \frac{\delta\mathrm{GF} - \lambda_{s_1}\,\delta N_{\mathrm{WL}}}{\lambda_{s_1} - \lambda_{s_2}}, \qquad \delta N_{s_1} \;=\; \delta N_{s_2} + \delta N_{\mathrm{WL}}, \tag{25}$$

where δGF is the in-arc TD2 or, across a data gap, the bridged jump of Section 1.4.4. The denominator $\lambda_{s_1} - \lambda_{s_2}$ (the variable `denom` in the source) is $\approx -0.0539\,\mathrm{m}$ for GPS L1/L2. The amplification factor on the wide-lane integer is therefore $\lambda_{s_1}/(\lambda_{s_1} - \lambda_{s_2}) \approx -3.53$. Despite that lever, the GF equation must localise δN_{s_2} to well within ± 0.5 cycle; the split-capability gate below enforces this.

1.4.2 Integer rounding and validation

The float δN_{s_2} from Equation (25) is rounded and its residual recorded:

$$d_2 = \mathrm{round}\big(\delta N_{s_2}\big), \quad f = \big|\delta N_{s_2} - d_2\big|, \quad d_1 = d_2 + \delta N_{\mathrm{WL}}. \tag{26}$$

A candidate is accepted as a repair only if three conditions hold. The rounding is clean ($f \le 0.30$, the constant `kMaxFractional`, which leaves a $1 - 0.30 = 0.70$ cycle margin to the nearest wrong integer). The magnitudes are sane ($|d_1|, |d_2| \le$ `kMaxSlipCycles`, which is 1000 cycles). The pair is non-trivial ($d_1 \neq 0 \vee d_2 \neq 0$). An accepted candidate yields `action="repair"`. The pair $(d_1, d_2) = (0, 0)$ resolves as no slip, and the arc continues. Any other outcome resets the satellite.

The joint solve is trusted only when the GF evidence is sharp enough to discriminate the integer lattice. The float δN_{s_2} carries $\sigma_{\mathrm{TD2}}/|\lambda_{s_1} - \lambda_{s_2}|$ cycles of noise, so the solve is permitted only when

$$\sigma_{\mathrm{TD2}} \;\le\; \texttt{turbo_edit_split_max_sigma} \cdot \big|\lambda_{s_1} - \lambda_{s_2}\big| \qquad (\texttt{splitCapable}), \tag{27}$$

Here σ_{TD2} is the arc's own MAD-derived σ when the arc is mature, otherwise the cross-satellite tier value, otherwise NaN. NaN fails the gate, because the absence of an estimate is no evidence of sharpness. The default of `estimation.cycle_slip.turbo_edit_split_max_sigma` is 0.2 cycles. This gate is the only mechanism that makes the resolver rate-adaptive. Millimetre-level noise at 1 s passes easily; the ionospheric residual at a coarse rate (30 s) passes only while the arc is quiet. An MW-flagged pair that fails the gate falls back to a satellite reset. A GF-only exceedance on an arc that fails the gate is ignored, not reset, and the Kalman residual gate remains the backstop. When $\delta N_{\mathrm{WL}} = 0$ (an equal slip $\delta N_{s_1} = \delta N_{s_2}$, to which MW is blind), a GF-only flag fires when |TD2| exceeds the adaptive threshold $\tau = \max(k_\sigma \hat{\sigma}_{\mathrm{TD2}}, 0.04\,\mathrm{m})$, with k_σ the `turbo_edit_sigma_multiplier` (default 6). The floor sits just below $|\lambda_{s_1} - \lambda_{s_2}| \approx 0.054\,\mathrm{m}$, so a ± 1-cycle equal slip remains detectable. The joint solve takes a zero wide-lane jump for a GF-only flag only when the MW detector affirmatively reports the pair clean this epoch. MW silence alone is not taken as a zero wide-lane jump: a locked or immature MW arc can miss a real unequal slip, and forcing zero would alias it into a confident wrong repair. Without that evidence the wide-lane integer is unknown and the satellite is reset. An unstable ionosphere on the constellation (the cross-satellite IVI above 0.06 m) also resets the satellite instead of solving.

1.4.3 MW-only / code-event null line

The asymmetric case is an MW flag with a GF jump that is affirmatively zero. For a fixed $\delta N_{\mathrm{WL}} = k$, every integer pair $(\delta N_{s_1}, \delta N_{s_1} - k)$ predicts, through Equation (22), the GF jump

$$\delta\mathrm{GF} \;=\; \lambda_{s_2} k - (\lambda_{s_2} - \lambda_{s_1})\, \delta N_{s_1}, \tag{28}$$

with its zero crossing at $\delta N_{s_1} = \lambda_{s_2} k/(\lambda_{s_2} - \lambda_{s_1}) \approx 4.53\, k$ for GPS L1/L2 ($\approx 3.95\, k$ for L1/L5). Every small pair with $|\delta N_{s_1}| \le 3$ then predicts $|\delta\mathrm{GF}| \ge 8\,\mathrm{cm}$, orders of magnitude above phase noise. A flat GF therefore leaves only *null-line* pairs of $(4, 3)$ / $(9, 7)$-class magnitude: multi-cycle simultaneous slips on both signals with no loss of lock, which are implausible. The simpler explanation is a narrow-lane code excursion (multipath, NLOS, DLL re-lock), since $\mathrm{MW} = \Phi_{\mathrm{WL}} - P_{\mathrm{NL}}$ and one wide-lane cycle corresponds to $\approx 0.86\,\mathrm{m}$ of narrow-lane code.

The pair is therefore classified as a code-domain artifact, and the caller skips the phase reset, when the GF jump is affirmatively zero within a gate bounded by the lattice discrimination:

$$|\mathrm{TD2}| \le \underbrace{\texttt{kCodeEventZeroSigma} \cdot \hat{\sigma}_{\mathrm{MAD}}}_{\text{zero gate}} \quad \wedge \quad \text{zero gate} \le \underbrace{\texttt{kVetoZeroGateMaxLambdaFrac} \cdot |\lambda_{s_1} - \lambda_{s_2}|}_{\text{discrimination cap}}, \tag{29}$$

with `kCodeEventZeroSigma` $= 3.5$ ($\approx 99.95\%$ two-sided) and `kVetoZeroGateMaxLambdaFrac` $= 0.5$. The cap is the rate-adaptive safeguard. A zero gate wider than half the smallest GF-visible quantum cannot distinguish stillness from a small slip; the veto then stands down and the MW flag governs. The classification has further trust conditions: the veto is enabled, MW flagged, GF did not exceed its threshold, no post-slip cooldown is active, the constellation's ionosphere is stable, neither signal carries an LLI, and the MAD maturity is ≥ 10 samples. The last condition is stricter than the 5 samples detection needs, because affirming stillness preserves state. The implied null-line pair is computed for the trace only and never applied; `did_slip` stays false so the GF chain continues. A real null-line slip is still caught. It moves that signal's Kalman phase innovation by $\ge |4\lambda_{s_1}| \approx 0.76\,\mathrm{m}$ in the same epoch, far beyond any residual gate, so the residual-driven reset corrects it one epoch late. The behaviour is toggled by `estimation.cycle_slip.turbo_edit_mw_only_veto` (default on).

1.4.4 Gap-spanning GF bridge

The TD2 chain needs three consecutive epochs. Any per-signal dropout or receiver gap re-seeds the chain, and the post-gap TD2 is then blind to a re-lock slip whose GF level jump falls between the two chains. When a chain break occurs within `estimation.cycle_slip.turbo_edit_max_gap_s`, the resolver stores the pre-gap GF level $\mathrm{GF}_{\mathrm{ref}}$, its timestamp, the arc's EWMA GF rate $\dot{\mathrm{GF}}$ and the pre-gap spacing dt_0. On return it forms the drift-corrected jump across the gap:

$$\boxed{\delta\mathrm{GF}_{\mathrm{gap}} \;=\; \mathrm{GF}_{\mathrm{now}} - \mathrm{GF}_{\mathrm{ref}} - \dot{\mathrm{GF}}\,\Delta t, \qquad \Delta t = t_{\mathrm{now}} - t_{\mathrm{ref}},} \tag{30}$$

which then takes the place of $\delta\mathrm{GF}$ in Equation (25). Here $\dot{\mathrm{GF}}$ is an EWMA ($\alpha = 0.1$) of the per-epoch GF rate, chosen so that its noise is not amplified when projected over the gap. The uncancelled ionospheric residual is a random walk over the gap. The detection threshold and the code-event zero gate are therefore widened by the factor $w = \sqrt{\Delta t/dt_0}$ (for $\Delta t > dt_0$), and the split-capability σ becomes $\sigma_{\mathrm{bridge}} = \hat{\sigma}_{\mathrm{MAD}}\, w$. The discrimination cap of the zero gate is not widened, because the integer lattice of Equation (23) does not stretch with the gap. The bridge cap defaults to 15 s in code and is raised to 60 s in the bundled real-time and replay configs.

An MW-flagged pair across the gap has three possible outcomes.

1. A *code event* (`action="code_event"`) when $\delta\mathrm{GF}_{\mathrm{gap}}$ is affirmatively zero within the widened gate. This is the null-line logic of Equation (29) applied across the gap, and it preserves phase through a re-acquisition DLL transient.

2. A *repair* or *moot* outcome when the joint solve runs with $\delta\mathrm{GF} = \delta\mathrm{GF}_{\mathrm{gap}}$. Each signal is dispatched by continuity in the observation domain. A signal observed at every epoch through the gap kept its ambiguity, so its nonzero δN is a *repair*. A signal that missed an epoch was pruned and reseeded on return, so its nonzero δN is *moot*: already absorbed, with the reset suppressed.

3. A *reset* when the across-gap σ is too coarse for the split, or when the candidate set holds no integer or more than one and no deferral applies.

The GF equation alone is degenerate along the null-line lattice of Equation (28): $+0.735\,\mathrm{m}$ fits both $(0, -3)$ and $(9, 4)$ at the millimetre level, and only the integer k separates them. The bridged solve therefore requires a unique surviving candidate after three guards. The first is the fractional gate of Equation (26). The second is a GF-separation guard: the candidate's own predicted jump must clear the zero gate, $|\lambda_{s_1} d_1 - \lambda_{s_2} d_2| >$ `zero_gate`, so that a marginal noise exceedance cannot produce a multi-cycle repair. The third is a sibling-pair zero affirmation. Another GF arc of the same satellite that stayed chain-continuous and unflagged through the bridged interval affirms $\delta N = 0$ for both its signals, and any candidate that needs a nonzero integer on an affirmed signal is pruned. The wide-lane integer for the bridged solve is taken in order of trust: the published MW integer k; failing that, at the flag epoch, $\mathrm{round}(f) \pm 3$ around the instantaneous re-acquisition float (biased by the narrow-lane code); and on deferral epochs the settled across-gap MW Hatch mean. When the float wide-lane evidence is unsettled or the lattice is ambiguous (two or more acceptable integers), the solve is deferred instead of guessed, up to `estimation.cycle_slip.turbo_edit_wl_settle_max_s` (default 45 s). The deferral is available only with the soft-LLI bridge (`turbo_edit_bridge_lli`), whose holds keep the parked phase measurements out of the filter meanwhile. A deferral ends in a later repair, a reset or a timeout reset.

1.4.5 Atomic reconciliation

Per-satellite candidates are reconciled atomically. A cross-pair integer disagreement on a shared signal demotes the whole satellite to reset. A satellite that ends in the reset set has its code-event, moot, clean and pending classifications withdrawn, so contradictory evidence always resolves to the conservative outcome. Arc bookkeeping maintains the TD2 precondition of consecutive epochs. A non-consecutive update partially resets the chain but keeps the slowly varying MAD calibration window. A gap beyond the arc's maximum clears the arc. The callers set that maximum to 1.5 times the data interval; the resolver's own default, 120 s, applies only when no caller sets it. An accepted slip re-seeds the chain with a cooldown of 3 epochs that suppresses the TD2 echo. Only clean TD2 samples (not exceeding the threshold, not bridged) feed the MAD window.

1.4.6 Soft-LLI hold

A receiver-declared loss of lock (LLI) normally resets the satellite's phase ambiguities and erases the geometry-free arc. When the gap is short and the data rate is at least 1 Hz on a dual-frequency satellite that is not a BSSD pivot or GLONASS, the soft-LLI bridge (`estimation.cycle_slip.turbo_edit_bridge_lli`, default off; `include/slip/SoftLliBridge.hpp`) holds the ambiguities instead. It keeps a rolling snapshot $(\hat{x}_{\mathrm{snap}}, \sigma^2_{\mathrm{snap}})$ of the converged ambiguity values and variances, taken after each update and only for keys below a variance ceiling. On an LLI it keeps the

geometry-free arc, excludes the held phase measurements from the update across the one-epoch MW confirmation delay, and lets the gap bridge of Section 1.4.4 decide the integer. A unique solve overwrites the ambiguity as

$$\hat{x} \;=\; \hat{x}_{\mathrm{snap}} + \delta N\,\lambda_s, \qquad \sigma^2 \;=\; \max\!\big(\sigma^2_{\mathrm{snap}},\; 10^{-6}\big), \tag{31}$$

with its cross-covariances set to zero, which keeps the converged variance of the ambiguity itself, and releases the hold. Otherwise the ambiguity is reset, and each held signal has its own timeout from its own hold start. The repair assumes that the carrier loop held phase coherence across the LLI. If it re-acquired, δN absorbs the re-acquisition and the restored ambiguity is still the current integer, but a wrong unique solve puts at least $|\delta N\,\lambda_s|$ into the phase innovation when the signal rejoins, and the residual gate resets it one epoch late. Both the post-processing and the real-time paths share this helper.

1.4.7 Non-causal two-sided bridge (staged)

`SlipBridgeKernel` and `JumpGapFitter` (`include/slip/`) recover the integer slip across a flagged boundary from samples on *both* sides, which a causal detector cannot use. They are compiled and tested, no production code calls them, and the four settings that name them are parsed and read nowhere. The fitter fits an independent line $y = a + b\,(t - t_b)$ to each side of the boundary t_b and reports the step of the two extrapolated levels at t_b. Applied to the geometry-free series and the MW series it gives the steps ΔGF and ΔMW, and with the signals sorted high frequency first the attribution is the algebra of Section 1.4.1:

$$\delta N_{\mathrm{WL}} = \mathrm{round}\big(\Delta\mathrm{MW}/\lambda_{\mathrm{WL}}\big), \quad \delta N_2 = \mathrm{round}\left(\frac{\Delta\mathrm{GF} - \lambda_1\,\delta N_{\mathrm{WL}}}{\lambda_1 - \lambda_2}\right), \quad \delta N_1 = \delta N_2 + \delta N_{\mathrm{WL}}. \tag{32}$$

The result is accepted when each rounding residual is at most 0.30 cycle and the integers are not both zero and do not exceed 1000 cycles in magnitude.

1.5 Response on the user path

The user path applies a slip in the filter state through two primitives in `src/core/slip/CycleSlipRepair.cpp`, driven by `CycleSlipManager::detectAndRepair` (`CycleSlipDetection.cpp`, `CycleSlipAdjudication.cpp`, `CycleSlipPivot.cpp`, `CycleSlipNonMwPasses.cpp`).

- *Repair* adds the integer jump to the ambiguity state, $\hat{x} \leftarrow \hat{x} + \delta N\,\lambda_s$, and leaves its covariance unchanged. A deterministic state jump is recorded for the RTS smoother as a control input, so the backward pass does not absorb it as an innovation. A repair is refused for a BSSD pivot, whose single-difference ambiguity is frozen at zero; the pivot case is handled by a variance inflation that makes the pivot manager re-reference the frame.
- *Reset* re-seeds the ambiguity with a large variance. With between-satellite differencing the seed is the single difference of the two absolute ambiguity seeds; when the pivot is not observed the variance is inflated and the re-seed deferred to the next epoch. The smoother segments the satellite at this epoch.

Which signal slipped follows from the cross-consistency of overlapping MW pairs on a satellite with three or more frequencies. A signal that appears in a flagged pair and in no unflagged pair is a culprit. With $\delta N_{\mathrm{WL}} = \delta N_{s_1} - \delta N_{s_2}$, a lone culprit s_1 takes the integer δN_{WL} and a lone culprit s_2 takes $-\delta N_{\mathrm{WL}}$. The integers gathered for one culprit across its pairs must agree; otherwise the signal

is reset. When all pairs are flagged there is no clean reference, the attribution is inconclusive, and the TurboEdit resolution of Section 1.4 or a full reset applies. The auxiliary detectors (GRAPHIC, EWL) trigger resets of the signals they flag after the MW path has handled its satellites.

References

[1] Simon Banville and Richard B. Langley. "Mitigating the impact of ionospheric cycle slips in GNSS observations". In: *Journal of Geodesy* 87.2 (2013), pp. 179–193. DOI: 10.1007/s00190-012-0604-1.

[2] Jianghui Geng and Yehuda Bock. "Triple-frequency GPS precise point positioning with rapid ambiguity resolution". In: *Journal of Geodesy* 87.5 (2013), pp. 449–460. DOI: 10.1007/s00190-013-0619-2.

[3] Thomas P. Yunck. "Coping with the atmosphere and ionosphere in precise satellite and ground positioning". In: *Environmental Effects on Spacecraft Positioning and Trajectories*. Geophysical Monograph Series 73 (1993), pp. 1–16. DOI: 10.1029/GM073p0001.